Camunda Event Listener

In this session we will look at what is is event listeners how it functions ?.

Event listeners are used to intercept process engine events.

Types of event listeners

1. Task Event Listeners
2. Execution Event Listeners.

1. Task Event listeners

A task listener is used to execute custom Java logic or an expression upon the occurrence of a certain task-related event.

Task events:

Create: Task create event fires when the task has been created and all task properties are set. No other task-related event will be fired before the *create* event. The event allows us to inspect all properties of the task when we receive it in the create listener.

Update: Task update event occurs when a task property (e.g. assignee, owner, priority, etc.) on an already created task is changed. This includes attributes of a task (e.g. assignee, owner, priority, etc.), as well as dependent entities (e.g. attachments, comments, task-local variables). Note that the initialization of a task does not fire an update event (the task is being created). This also means that the update event will always occur after a create event has already occurred.

Assignment: Task assignment event specifically tracks the changes of the Task's assignee property. The event may be fired on two occasions:

1. When a task with an assignee explicitly defined in the process definition has been created. In this case, the *assignment* event will be fired after the *create* event.

2. When an already created task is assigned, i.e. the Task's assignee property is changed. In this case, the *assignment* event will follow the *update* event since changing the assignee property results in an updated task.

Timeout: The timeout event occurs when a Timer, associated with this Task Listener, is due. Note that this requires for a Timer to be defined. The timeout event may occur after a Task has been created, and before it has been completed.

Complete: Task event complete event occurs when the task is successfully completed and just before the task is deleted from the runtime data. A successful execution of a task's complete Task Listeners results in an end of the task event lifecycle.

Delete: The delete event occurs just before the task is deleted from the runtime data

1. An interrupting Boundary Event;
2. An interrupting Event Subprocess;
3. A Process Instance deletion;
4. A BPMN Error thrown inside a Task Listener.

No other event is fired after the *delete* event since it results in an end of the task event lifecycle.

2. Execution Listeners

Execution listeners allow you to execute external Java code or evaluate an expression when certain events occur during process execution. The events that can be captured are:

- Start and end of a process instance.
- Taking a transition.
- Start and end of an activity.
- Start and end of a gateway.

- Start and end of intermediate events.
- Ending a start event or starting an end event.

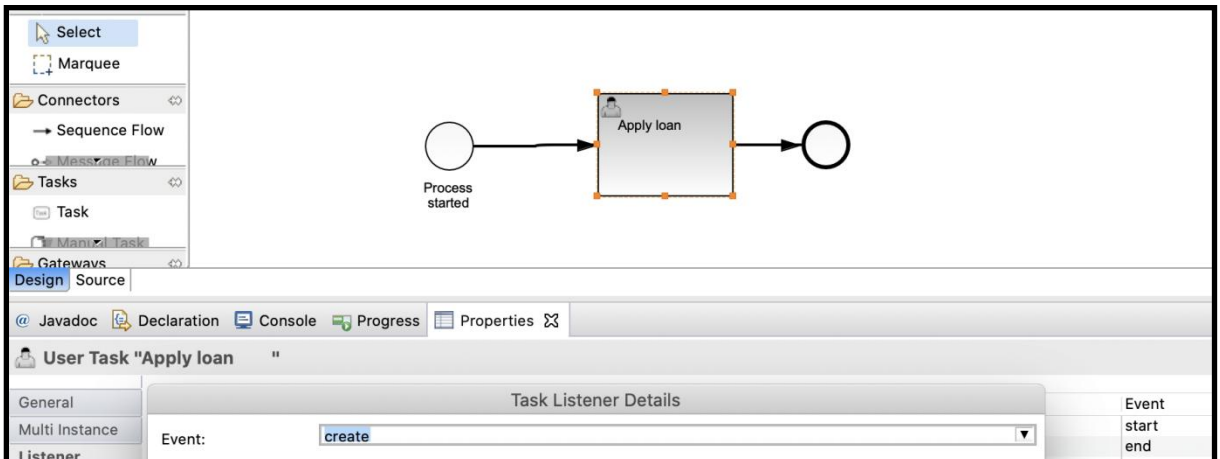
Let's demonstrate the task event listener

1. Define task listener CustomTaskEventListener.java

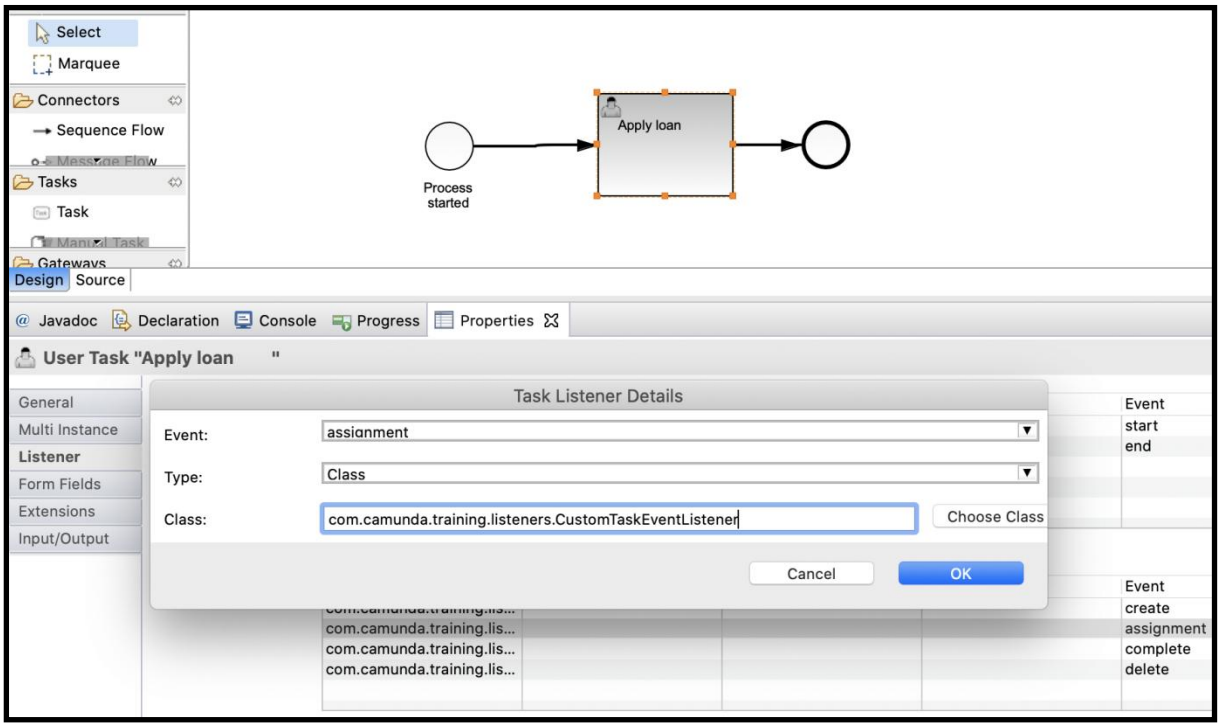
```
package com.camunda.training.listeners;

import java.util.logging.Logger;

import org.camunda.bpm.engine.delegate.DelegateTask;
import org.camunda.bpm.engine.delegate.TaskListener;

public class CustomTaskEventListener implements TaskListener {

	private final Logger LOGGER =
Logger.getLogger(CustomTaskEventListener.class.getName());

	@Override
	public void notify(DelegateTask delegateTask) {
		LOGGER.info("Task event occured : " +
delegateTask.getEventName());
		String firstName = (String) delegateTask.getVariable("firstName");
		String lastName = (String) delegateTask.getVariable("lastName");
		Long loanAmount = (Long) delegateTask.getVariable("loanAmount");

		LOGGER.info("First name : " + firstName);
		LOGGER.info("Last name : " + lastName);
		LOGGER.info("Loan amount : " + loanAmount);

	}

}
```

2. Configure user task to intercept task event

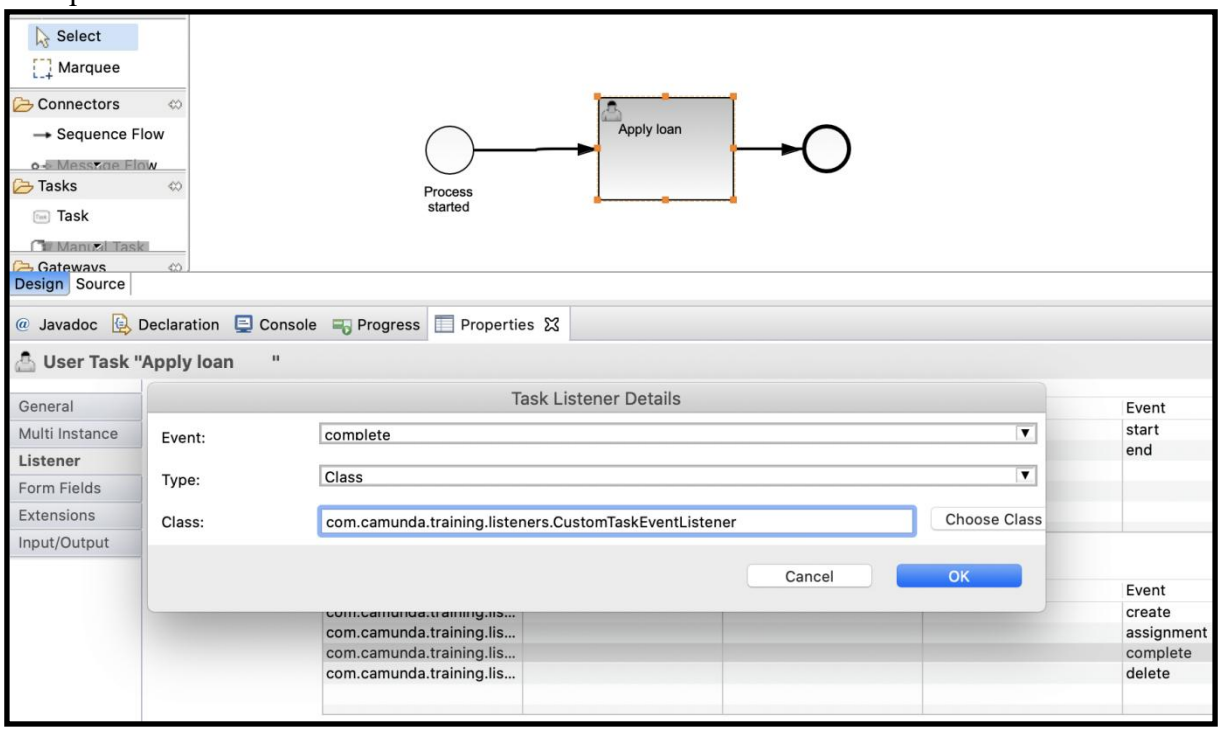
Create event:



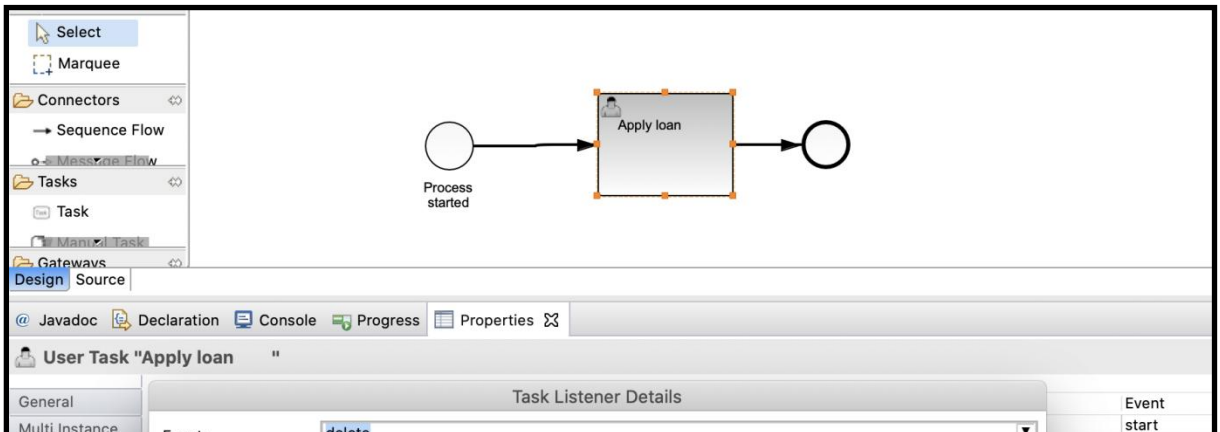
Assignment event:

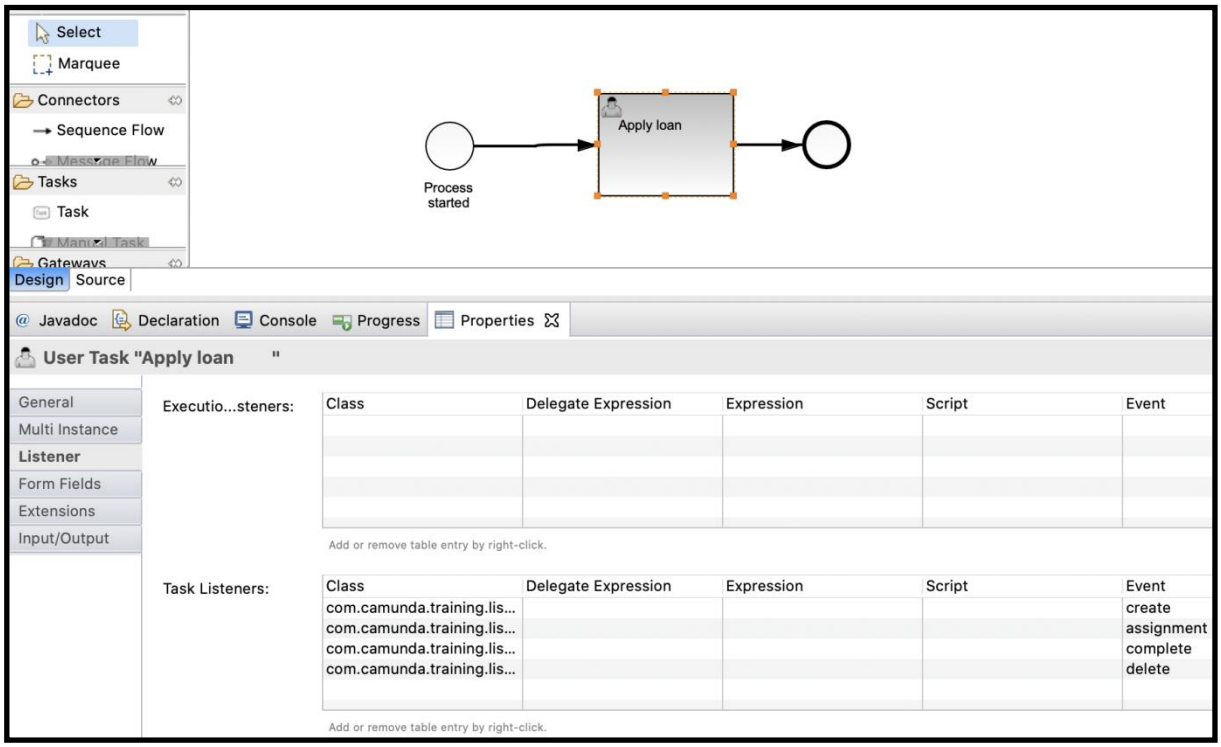


Complete event:



Delete event:

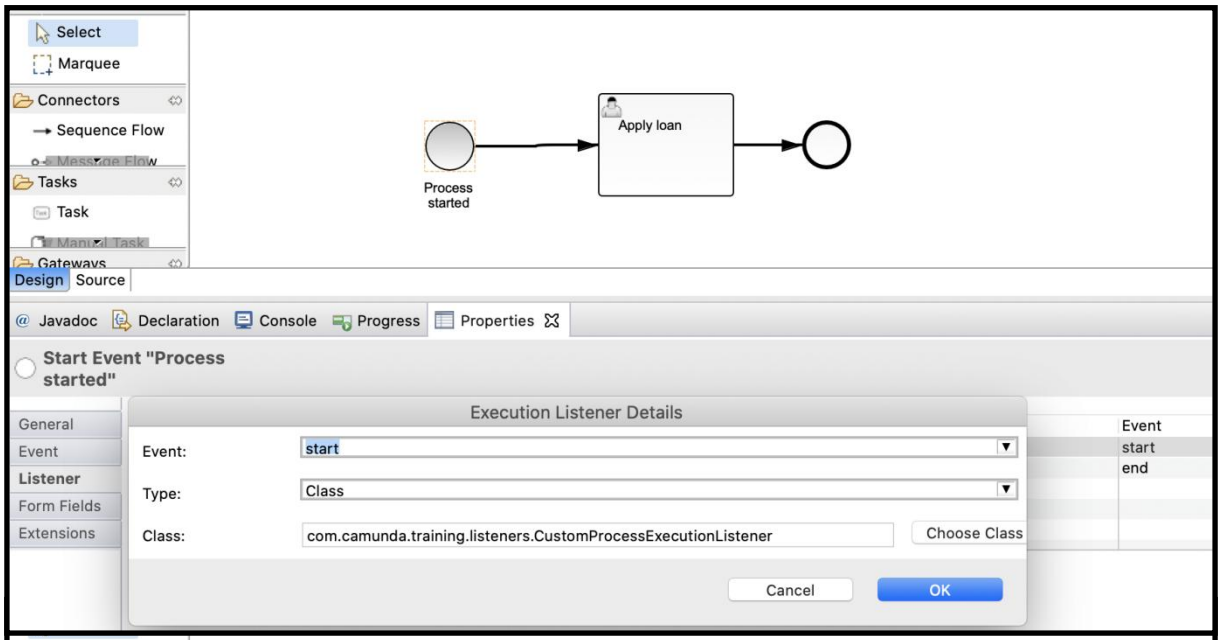


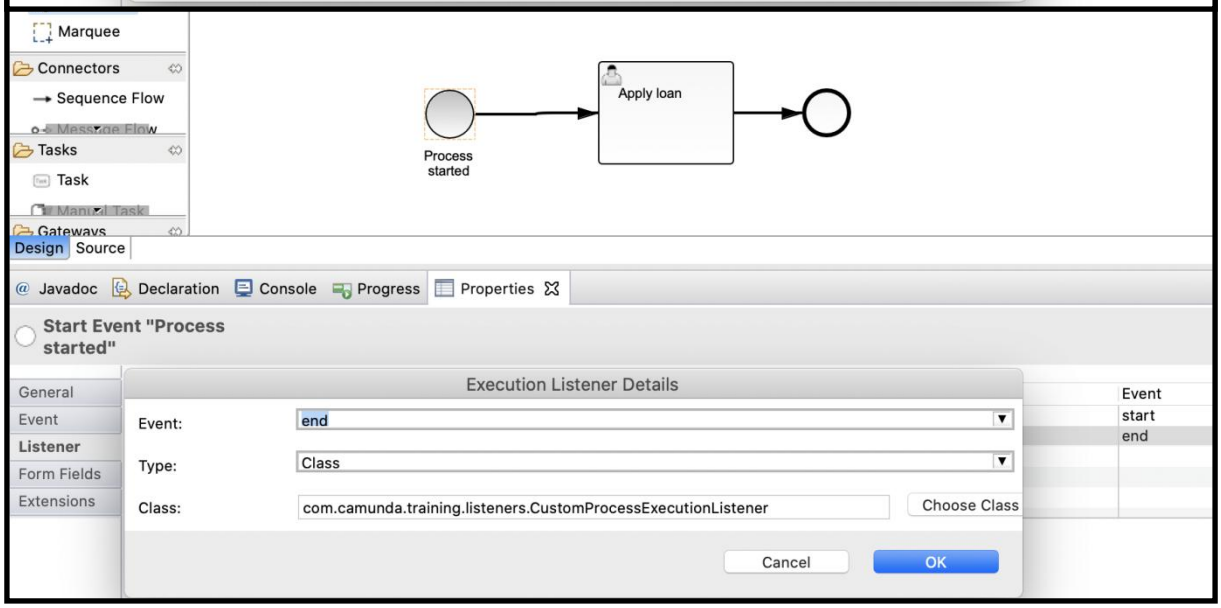


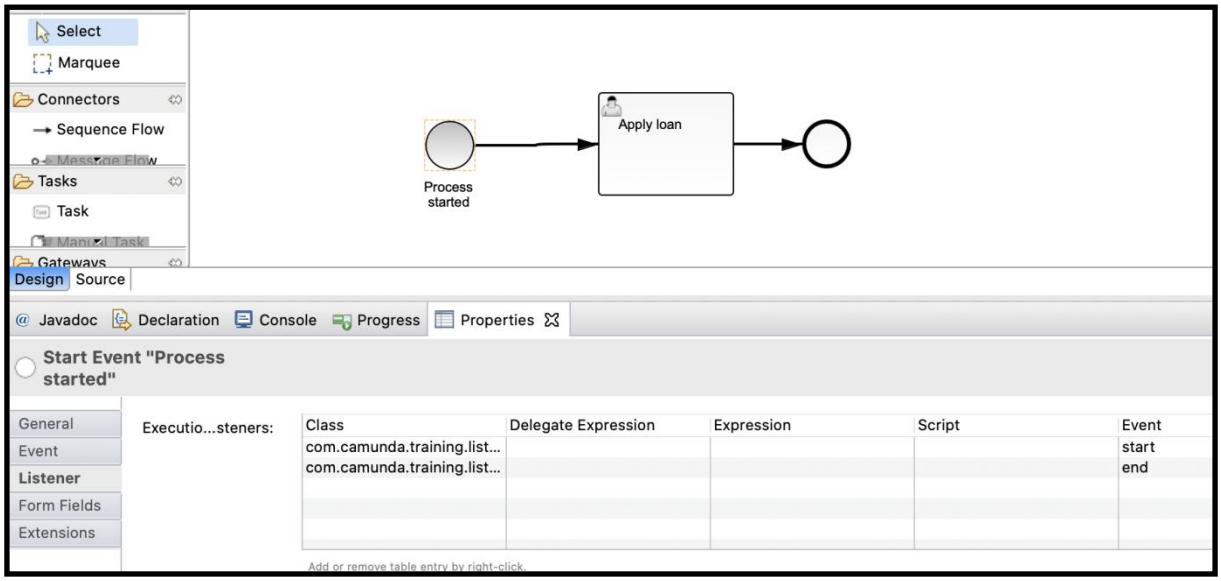
Task listener added for all the event.

Let’s demonstrate the execution event listener

Start event:

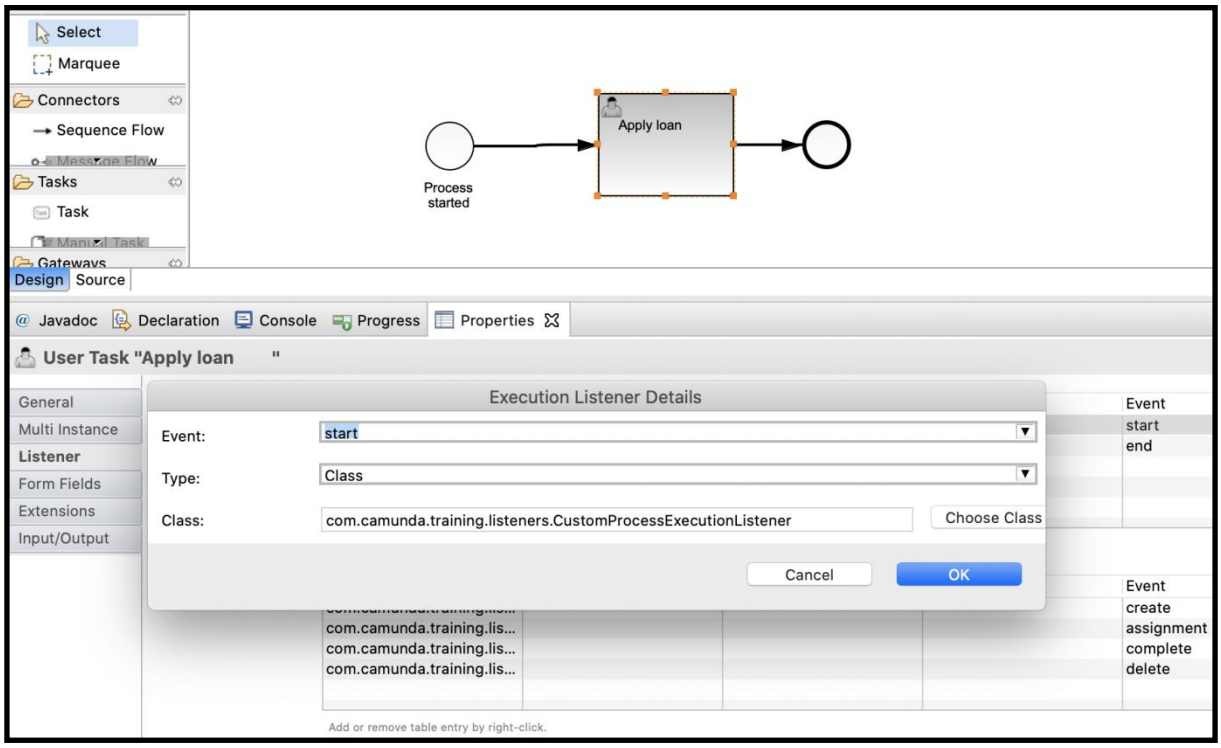


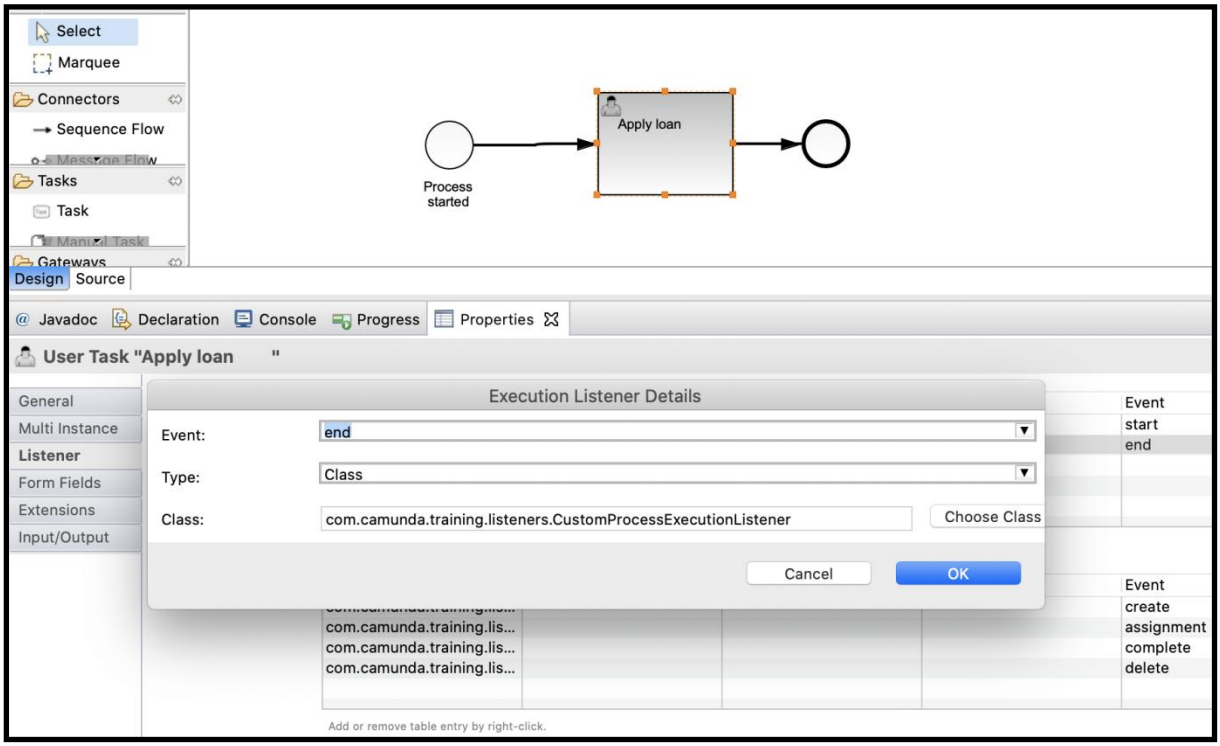




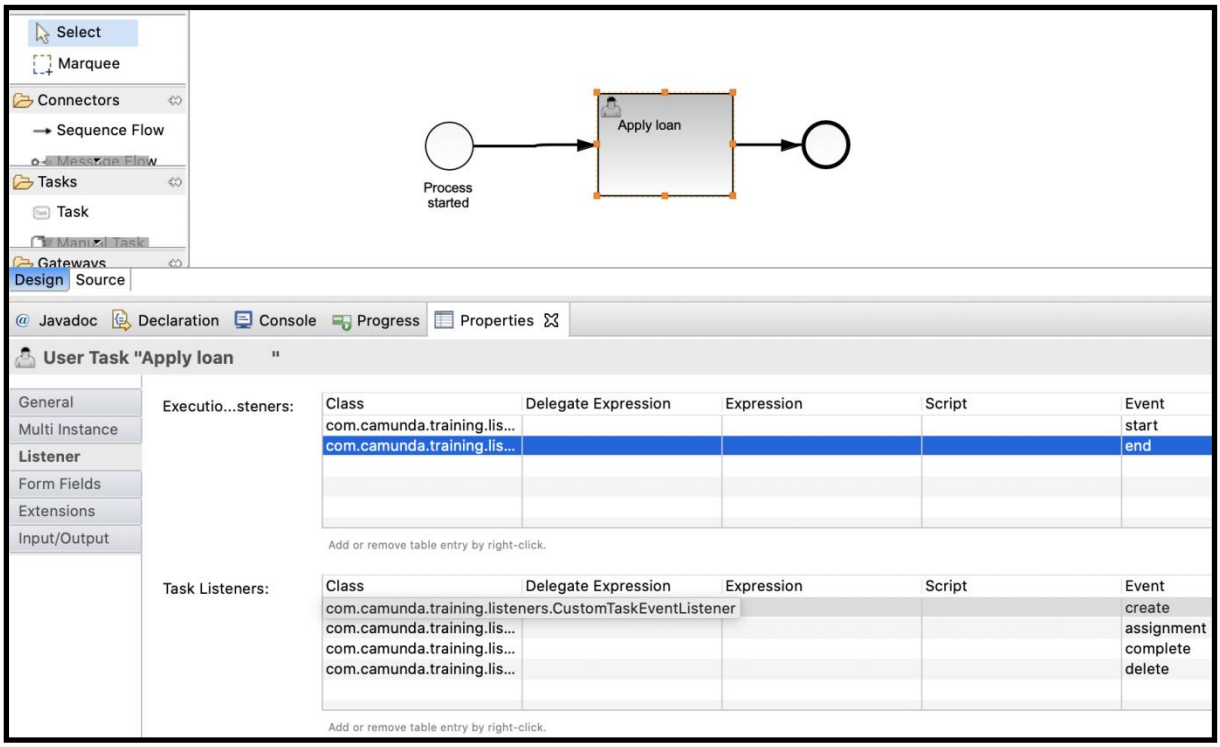
Execution listeners added to start event:

Task execution listener

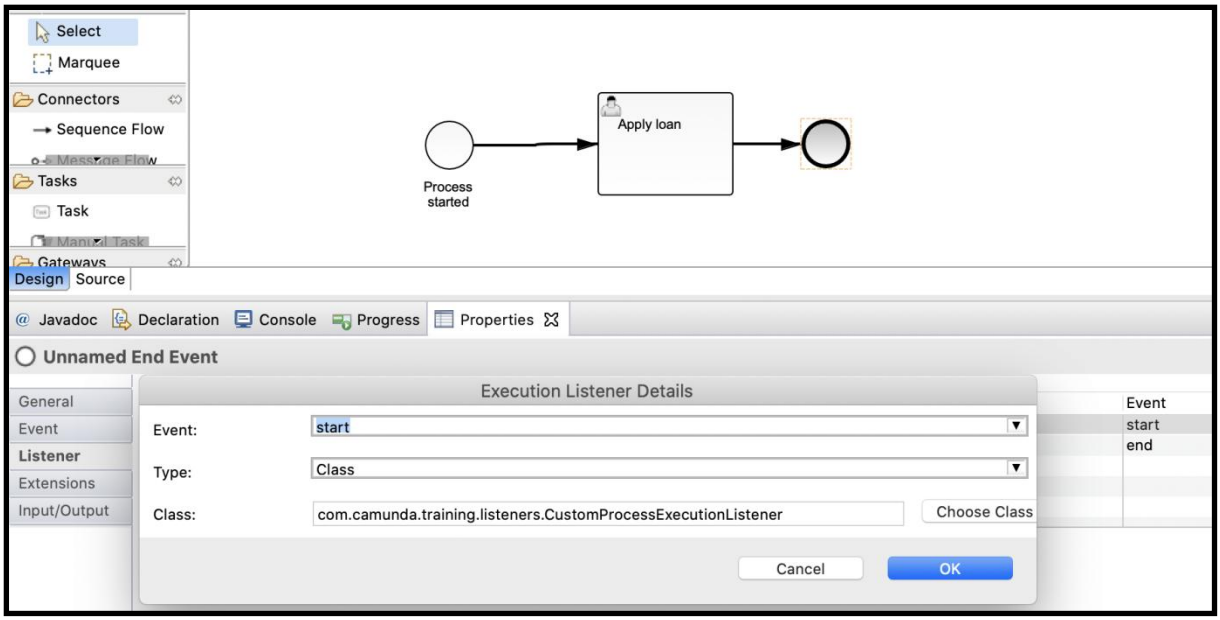


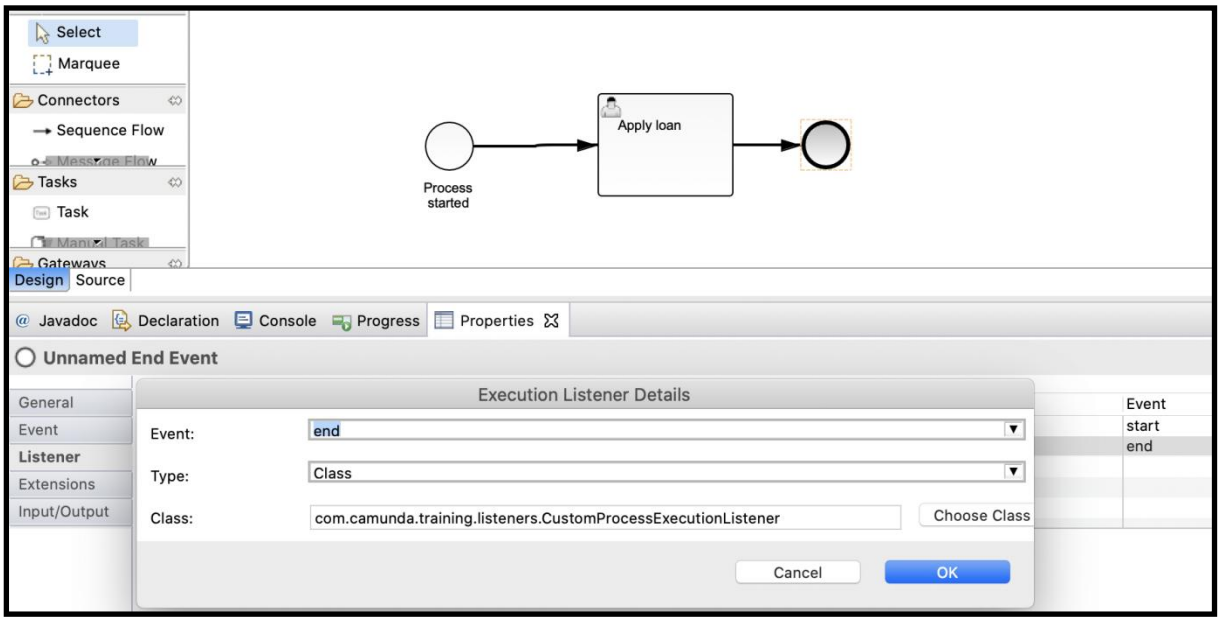


All the task execution listeners



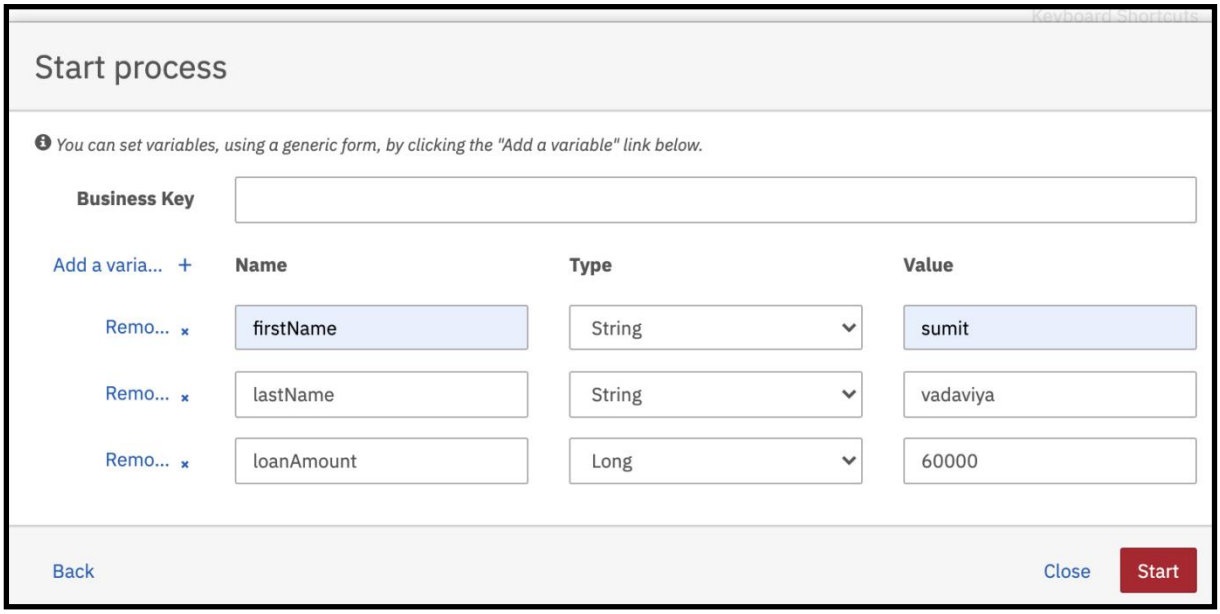
End event execution listener





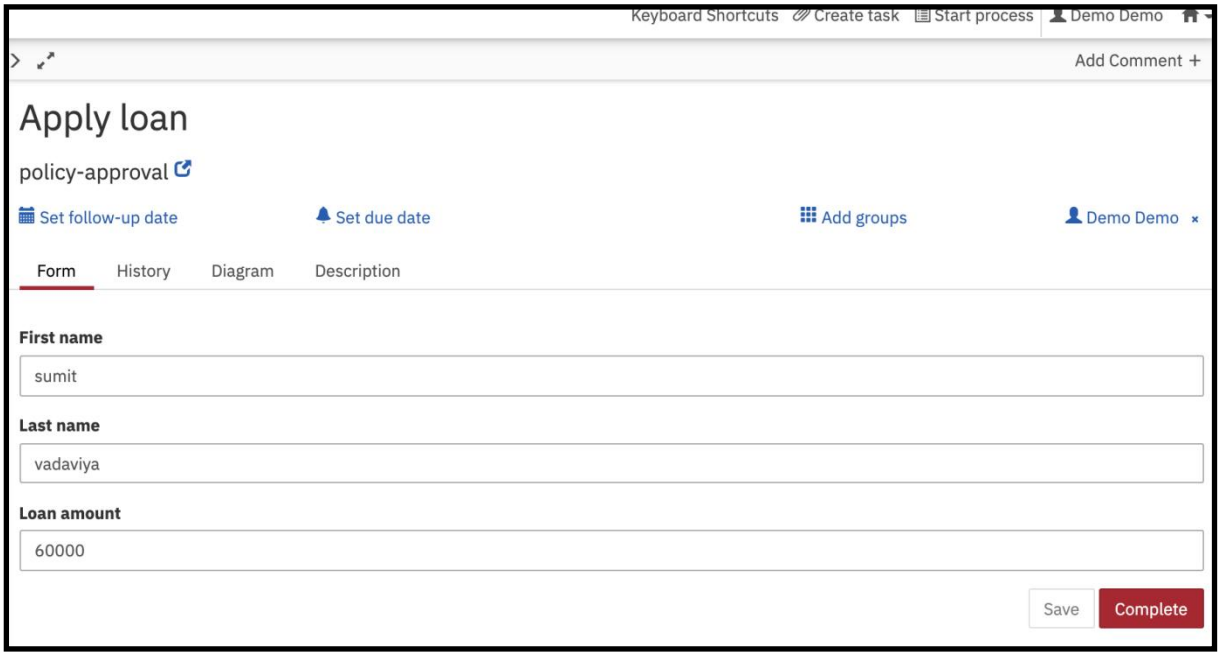
Deploy the application process and start the process.

Hit the policy-approval process.



Hit the start button to start the process.

As per process definition, Apply load user task will be created



Hit the Complete button.

Observe tomcat console log to verify task event listener and execution event listener executed.

05-Sep-2020 23:39:59.268 INFO [http-nio-8081-exec-23] com.camunda.training.listeners.CustomProcessExecutionListener.notify Task event occured : start
05-Sep-2020 23:39:59.268 INFO [http-nio-8081-exec-23] com.camunda.training.listeners.CustomProcessExecutionListener.notify First name : null
05-Sep-2020 23:39:59.268 INFO [http-nio-8081-exec-23] com.camunda.training.listeners.CustomProcessExecutionListener.notify Last name : null
05-Sep-2020 23:39:59.268 INFO [http-nio-8081-exec-23] com.camunda.training.listeners.CustomProcessExecutionListener.notify Loan amount : null
05-Sep-2020 23:39:59.268 INFO [http-nio-8081-exec-23] com.camunda.training.listeners.CustomProcessExecutionListener.notify Task event occured : end

```
05-Sep-2020 23:39:59.268 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomProcessExecutionListener.notify First
name : null
05-Sep-2020 23:39:59.268 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Last
name : null
05-Sep-2020 23:39:59.268 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Loan
amount : null
05-Sep-2020 23:39:59.269 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Task
event occured : start
05-Sep-2020 23:39:59.269 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomProcessExecutionListener.notify First
name : sumit
05-Sep-2020 23:39:59.269 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Last
name : vadaviya
05-Sep-2020 23:39:59.269 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Loan
amount : 60000
05-Sep-2020 23:39:59.270 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomTaskEventListener.notify Task event
occured : create
05-Sep-2020 23:39:59.271 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomTaskEventListener.notify First name :
sumit
05-Sep-2020 23:39:59.271 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomTaskEventListener.notify Last name :
vadaviya
05-Sep-2020 23:39:59.271 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomTaskEventListener.notify Loan amount :
60000
05-Sep-2020 23:39:59.271 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomTaskEventListener.notify Task event
occured : assignment
05-Sep-2020 23:39:59.272 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomTaskEventListener.notify First name :
sumit
05-Sep-2020 23:39:59.272 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomTaskEventListener.notify Last name :
vadaviya
```

```
05-Sep-2020 23:39:59.272 INFO [http-nio-8081-exec-23]
com.camunda.training.listeners.CustomTaskEventListener.notify Loan amount :
60000
05-Sep-2020 23:41:32.984 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomTaskEventListener.notify Task event
occured : complete
05-Sep-2020 23:41:32.985 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomTaskEventListener.notify First name :
sumit
05-Sep-2020 23:41:32.985 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomTaskEventListener.notify Last name :
vadaviya
05-Sep-2020 23:41:32.985 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomTaskEventListener.notify Loan amount :
60000
05-Sep-2020 23:41:32.987 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Task
event occured : end
05-Sep-2020 23:41:32.987 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify First
name : sumit
05-Sep-2020 23:41:32.987 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Last
name : vadaviya
05-Sep-2020 23:41:32.987 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Loan
amount : 60000
05-Sep-2020 23:41:32.989 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Task
event occured : start
05-Sep-2020 23:41:32.989 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify First
name : sumit
05-Sep-2020 23:41:32.989 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Last
name : vadaviya
05-Sep-2020 23:41:32.989 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Loan
amount : 60000
05-Sep-2020 23:41:32.989 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Task
event occured : end
```

```
05-Sep-2020 23:41:32.989 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify First
name : sumit
05-Sep-2020 23:41:32.989 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Last
name : vadaviya
05-Sep-2020 23:41:32.990 INFO [http-nio-8081-exec-30]
com.camunda.training.listeners.CustomProcessExecutionListener.notify Loan
amount : 60000
```